CHAPTER 9

RAY OPTICS AND OPTICAL INSTRUMENTS

CHAPTER ESSENCE AT A GLANCE:
Light propagation through reflection and refraction, total internal reflection, lens maker's formula, prism dispersion, and optical instruments.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Total Internal Reflection:** Light striking a rarer boundary at an incidence angle greater than the critical angle reflects completely back into the denser medium.

• **Lens Maker's Relation:** Focal length of an optical lens is determined by relative refractive index and radii of curvature of both refracting surfaces.

• **Prism Refraction & Minimum Deviation:** Light bends toward the base; at minimum deviation, rays pass symmetrically through the prism with equal incidence and emergence angles.

• **Compound Microscope & Telescope:** Microscope provides high two-stage magnification for small near objects; astronomical telescope employs large aperture objective to resolve distant celestial bodies.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Mirror formula for spherical mirrors

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Lens Maker's formula for thin spherical lenses

$$m = \frac{L}{f_0}\left(1 + \frac{D}{f_e}\right)$$

Magnifying power of compound microscope

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In ray optics derivations, missing direction arrowheads on light rays costs 1 Full Mark automatically!

• **Alert:** [Valuation Rubric] Draw neat ray diagram with proper direction arrowheads, lens centers, and foci .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Laws of refraction, Snell's law, and absolute refractive index [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Laws of Refraction & Snell's Law]** 1. First Law: The incident ray, refracted ray, and the normal to the refracting surface at the point of incidence all lie in the same plane. [1 Mark]

• 2. Snell's Law: The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for a given pair of media and given color of light: sin i divided by sin r = n_{21} = n_2 divided by n_1. In symmetrical form: n_1 sin i = n_2 sin r. Absolute refractive index: n = c divided by v (where c is speed in vacuum, v in medium). [1 Mark]

• **Key Exam Result:** n_1 sin i = n_2 sin r; Absolute Refractive Index n = c divided by v

2. Total Internal Reflection (TIR): condition, critical angle formula sin i c = 1 divided by n [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[TIR Concept & Two Conditions]** When light travels from an optically denser medium to a rarer medium, beyond a certain incident angle, the ray is completely reflected back into the denser medium. Conditions: 1. Light must travel from optically denser to rarer medium. 2. Angle of incidence must be greater than critical angle (i > i c). [1 Mark]

• **[Critical Angle Formula]** At critical angle i = i c, angle of refraction r = 90°. Applying Snell's law: n_1 sin i c = n_2 sin 90° = n_2 (1) implies sin i c = n_2 divided by n_1. When rarer medium is air (n_2 = 1, n_1 = n): sin i c = 1 divided by n implies i c = $\sin^{-1}$(1 divided by n). [2 Marks]

• **Key Exam Result:** sin i c = 1 divided by n; For glass (n=1.5), i c approx. 41.8°; For diamond (n=2.42), i c approx. 24.4°

3. Applications of TIR: optical fibers, mirage, and totally reflecting prisms [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Three Applications of TIR]** 1. Optical Fibers: Consist of a high-index glass core (n_1 approx. 1.5) surrounded by lower-index cladding (n_2 approx. 1.48). Light entering at small angle strikes core-cladding boundary at i > i c, undergoing multiple successive total internal reflections without attenuation over kilometers. Used in high-speed optical broadband and endoscopy. [1 Mark]

• 2. Mirage: In deserts on hot days, air near ground is hotter and optically rarer than upper cooler air. Light from sky or trees bends progressively away from normal until i > i c, undergoing TIR and reaching observer's eye, creating inverted image resembling a reflection on water surface. [1 Mark]

• 3. Totally Reflecting Prisms: 45°-90°-45° glass prisms (critical angle ~42°) deflect light rays by 90° or 180° by TIR without any silvering coating, used in periscopes and binoculars. [1 Mark]

• **Key Exam Result:** Applications: Optical fibers (telecom or endoscopy), Mirage, Totally reflecting prisms

4. Thin lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ and magnification [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Thin Lens Formula]** From Lens Maker's derivation: 1 divided by v - 1 divided by u = (n - 1)(1 divided by R_1 - 1 divided by R_2) and 1 divided by f = (n - 1)(1 divided by R_1 - 1 divided by R_2). Equating both expressions gives the Thin Lens Formula: 1 divided by f = 1 divided by v - 1 divided by u. [1 Mark]

• **[Linear Magnification (m)]** Ratio of image height h' to object height h: m = h' divided by h = v divided by u = f divided by (f + u) = (f - v) divided by f. For convex lens: m is negative for real-inverted image; positive for virtual-erect image. [1 Mark]

• **Key Exam Result:** 1 divided by f = 1 divided by v - 1 divided by u; Magnification m = v divided by u = h' divided by h

5. Combination of two thin lenses in contact: equivalent focal length and power [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Derivation]** Two thin lenses L_1 and L_2 of focal lengths f_1 and f_2 placed in contact coaxially. Point object O placed at distance u from first lens. [1 Mark]

• First lens produces image at v_1: 1 divided by v_1 - 1 divided by u = 1 divided by f_1. (Eq 1) Image at v_1 acts as virtual object for second lens, producing final image at v: 1 divided by v - 1 divided by v_1 = 1 divided by f_2. (Eq 2) [1 Mark]

• **[Combination Formula]** Adding Eq 1 and Eq 2: 1 divided by v - 1 divided by u = 1 divided by f_1 + 1 divided by f_2. If equivalent focal length is F, then 1 divided by v - 1 divided by u = 1 divided by F. Therefore: 1 divided by F = 1 divided by f_1 + 1 divided by f_2. Power of lens P = 1 divided by f (in meters, unit: Dioptre D). Total power of combination: P = P_1 + P_2. [1 Mark]

• **Key Exam Result:** 1 divided by F = 1 divided by f_1 + 1 divided by f_2; Total Power P = P_1 + P_2 (in Dioptres)

6. Numerical on Lens Maker's formula with liquid immersion [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formulas in Air and Liquid]** In air: 1 divided by f a = (n g - 1) [1 divided by R_1 - 1 divided by R_2]. In liquid of refractive index n l: 1 divided by f l = (n g divided by n l - 1) [1 divided by R_1 - 1 divided by R_2]. Dividing both equations: f l divided by f a = (n g - 1) divided by (n g divided by n l - 1). [1.5 Marks]

• **[Step 2: Substitution & Evaluation]** Given f a = 20 cm, n g = 1.5, liquid is water n l = 4 divided by 3 = 1.333. f l divided by 20 = (1.5 - 1) divided by (1.5 divided by 1.333 - 1) = 0.5 divided by (1.125 - 1) = 0.5 divided by 0.125 = 4. f l = 20 × 4 = 80 cm. Focal length quadruples when immersed in water! [1.5 Marks]

• **Key Exam Result:** Focal length in water f l = 80 cm (increases 4 times)

7. Numerical on critical angle and total internal reflection [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula]** Critical angle i c = $\sin^{-1}$(1 divided by n). [1 Mark]

• **[Step 2: Substitution & Problem]** For glass-air interface with refractive index n = 1.5: sin i c = 1 divided by 1.5 = two-thirds = 0.6667 implies i c = $\sin^{-1}$(0.6667) = 41.8°. For diamond with n = 2.42: sin i c = half (0.5).42 = 0.4132 implies i c = 24.4°. Because of this very small critical angle, a diamond cut with proper facets causes multiple internal reflections, giving brilliant sparkling luster. [1 Mark]

• **Key Exam Result:** Critical angle for glass: 41.8°; Critical angle for diamond: 24.4°